THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest EVEN number that can be formed by using each of the digits 3, 1, 4, 2 once is: 1

(a) 1324 **(b)** 4312 **(c)** 1234 **(d)** 1342

2. The number of 3-digit palindromes is: 1

(a) 99 **(b)** 81 **(c)** 90 **(d)** 100

3. The greatest 5-digit palindrome with first digit 2 is: 1

(a) 99999 **(b)** 2992 **(c)** 29992 **(d)** 29892

4. Starting from 16, the hailstone (Collatz) sequence reaches 1 in how many steps? 1

(a) 4 **(b)** 5 **(c)** 6 **(d)** 3

5. The next palindrome after 1331 is: 1

(a) 1221 **(b)** 1342 **(c)** 1441 **(d)** 1551

6. The sum of the first 30 counting numbers $1 + 2 + 3 + \cdots + 30$ is: 1

(a) 465 **(b)** 900 **(c)** 930 **(d)** 495

7. A Kaprekar step on 6174 gives: 1

(a) 6264 **(b)** 6174 **(c)** 6165 **(d)** 6183

8. The greatest 4-digit number whose digit sum is 10 is: 1

(a) 5500 **(b)** 9010 **(c)** 8200 **(d)** 9100

9. The smallest EVEN number that can be formed by using each of the digits 5, 6, 7, 2 once is: 1

(a) 5276 **(b)** 2756 **(c)** 2576 **(d)** 7652

10. The greatest 5-digit palindrome with first digit 4 is: 1

(a) 49894 **(b)** 99999 **(c)** 49994 **(d)** 4994

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the greatest and the smallest 5-digit numbers formed from 4, 9, 1, 7, 3, each digit once. 2

12. Apply the reverse-and-add steps on 87 and state the palindrome reached. 2

13. In a supercell pyramid each number is the sum of the two numbers just below it. The bottom row is 2, 3, 1, 4. Find the top number. 2

14. Find the greatest and the smallest 5-digit numbers formed from 8, 2, 6, 0, 5, each digit once. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Perform two Kaprekar steps on 1795. 3

16. Find the largest palindrome less than 1300 and the smallest palindrome greater than 1300. 3

17. Find the sum of all numbers from 1 to 40 by pairing the first and last numbers. 3

OR

Perform two Kaprekar steps on 2863. 3

18. Find the largest palindrome less than 2450 and the smallest palindrome greater than 2450. 3

OR

Find the sum of all numbers from 1 to 60 by pairing the first and last numbers. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A 4-digit palindrome has first digit 2 and digit sum 10. Find the number. Add 11 to it and see if the answer is a palindrome. 5

OR

Using the digits 3, 7, 5, 1 once each, form a 4-digit number closest to 5000. Show the options you tried and the difference. 5

20. A 4-digit palindrome has first digit 3 and digit sum 14. Find the number. Add 11 to it and see if the answer is a palindrome. 5

OR

Using the digits 8, 2, 6, 4 once each, form a 4-digit number closest to 6000. Show the options you tried and the difference. 5

END OF QUESTION PAPER — Total: 40 Marks

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